

AquaNeural

Autonomous Intelligence for Aquaculture

A pre-development business case for an enterprise IoT and AI-enabled aquaculture management system.

ROLE	STATUS	DOMAIN
Co-Project Manager IoT / Technology Lead	Concept / Pre-Development	Aquaculture • IoT • AI Business Transformation

PROJECT SNAPSHOT

AquaNeural proposes a transition from fragmented manual pond monitoring to unified, data-driven precision aquaculture. The business case covers problem definition, strategic alignment, technology migration, financial analysis, market sizing, alternatives, risks and approval.

#10M	4	8	Q3 2026-Q4 2027
Funding ask	Migration phases	Identified risks	Planned timeline

Prepared by Greatness Marcus

Project Management • Business Analysis • Technology Project Planning

01 | BUSINESS PROBLEM

From operational friction to a scalable control system

The business case identifies three connected weaknesses in enterprise-scale aquaculture: limited human oversight across many ponds, blind or fixed-schedule feeding, and unreliable manual data. Together, these create operational, financial and reporting risk.

01	Unscalable oversight Critical water-parameter events can occur outside effective human coverage across large pond networks.
02	Systemic feeding failure Fixed-schedule feeding can contribute to simultaneous water-quality problems across multiple ponds.
03	Data unreliability Manual testing creates gaps and records that are difficult to use for auditable, investor-grade reporting.

PROPOSED SOLUTION

Sense	Distributed sensors monitor dissolved oxygen, pH, temperature and ammonia across active ponds.
Decide	Edge computing and an enterprise dashboard process signals and surface health scores, alerts and trends.
Act	Actuators such as aerators, dosing pumps and automated feeders support rapid intervention when conditions require it.
Report	Timestamped operational data supports performance visibility and investor-ready reporting.

EXPECTED BUSINESS VALUE

- Improve visibility across multiple ponds through continuous monitoring.
- Support data-driven feeding and environmental intervention.
- Create structured, timestamped operational records.
- Reduce dependence on manual observation and reactive firefighting.
- Provide a foundation for scalable enterprise aquaculture management.

02 | PROJECT MANAGEMENT

Turning the concept into a delivery roadmap

My contribution combined project coordination with technology-focused analysis. The project team structured the business case around defined roles, delivery phases, financial assumptions, strategic fit, alternatives and risk controls.

Co-Project Manager	Co-led the business case development process, coordinated team activities and owned project deliverables alongside the second Co-Project Manager.
IoT / Technology Lead	Contributed to technical feasibility, system architecture, hardware requirements and implementation constraints.

DELIVERY ROADMAP

PHASE 0 Q3 2026	Site assessment & enterprise readiness
PHASE I Q4 2026	Enterprise hardware installation and calibration
PHASE II Q1 2027	Software, AI & enterprise dashboard deployment
PHASE III Q2-Q3 2027	Training, validation, full cutover & live operation

GOVERNANCE & CONTROL

- Six defined go/no-go criteria govern the transition to scale-out.
- Manual monitoring remains operational until validation and cutover requirements are satisfied.
- Eight project risks are assessed with likelihood, impact and mitigation strategies.
- Enterprise training covers dashboard interpretation, alert response, sensor maintenance and manual override procedures.

CORE PM METHODS DEMONSTRATED

Business Case	WBS / Planning	Risk Management	Stakeholders
Financial Analysis	Phased Delivery	Change / Transition	Technology Planning

03 | BUSINESS & FINANCIAL CASE

A project framed as an investment decision

The business case evaluates the proposed investment, revenue model, market opportunity and downside scenarios so the project can be assessed as a business decision.

Funding ask	Initial investment	Subscription	Year 2 EBITDA
₦10M	₦8.3M	₦250K / farm / month	₦9.9M

The model identifies Year 1 as the planned investment year, with EBITDA turning positive in Year 2 and cumulative break-even reached in early Year 2 under the base case.

MARKET SIZING

TAM	~12,000 farms	₦36B / year
SAM	~2,000 farms	₦6B / year
SOM	5 farms by Year 3; 15 by Year 5	₦45M / year

SCENARIO THINKING

- Base: Year 1 EBITDA (₦6.6M); Year 2 ₦9.9M; Year 3 ₦15.6M.
- Optimistic: Year 2 ₦13.1M; Year 3 ₦20.4M.
- Pessimistic: Year 2 ₦4.8M; Year 3 ₦9.2M.
- Worst case: hardware costs +30%; Year 2 ₦1.6M; Year 3 ₦6.8M.

PORTFOLIO TAKEAWAY

AquaNeural demonstrates my ability to connect project management with business analysis and technology. The work combines problem definition, phased delivery planning, risk thinking, financial modelling, market sizing and technology implementation considerations in a single project case.

SKILLS DEMONSTRATED

Project Management	Business Analysis	Risk Management
Business Case Development	Strategic Planning	IoT / Technology Planning

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